

Gene Pulser® Electroporation

Cell Type Bacterial, gram positive

Molecules Electroporated DNA: pMU1328, pSA3, pGB301, pJDC9, pAMB1; 3 to 30kB, primarily covalently closed circular form.

Species Used *Lactococcus lactis*, subspecies *cremoris & lactis*, LM0230, H2, HP

Before the Pulse

Cell Growth Medium M17 glucose broth (ATCC)

Growth Phase at Harvest O.D. (600) = 0.3 to 0.7

Pre-pulse Incubation ice

Wash Solution water

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C (ice)

Electroporation Medium water

Cuvette Gap 0.2 cm

Voltage 2 to 2.5 kV

Cell Density 10 (10) x 10 (11) cells / ml

Volume of Cells 40 µl

Field Strength 10 to 12.5 kV/cm

DNA Concentration 100 to 500 ng in 1 to 10 µl

DNA Resuspension Buffer water or TE Buffer (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 25 µF

Volume of DNA 1 to 10 µl

Resistor (Pulse Controller) 200 Ω

After the Pulse

Time Constant 3.9 to 4.7 msec

Outgrowth Medium M17 glucose+0.5 M sucrose broth

Relevant Publications and/or Comments

Note: exponential values designated in parentheses. Powell *et al.*, 1988. *Appl. Environ. Microbiol.* **54**:655-660.

Outgrowth Temperature 30 °C

Length of Incubation 1 hour

Selection Method or Assay Used antibiotic marker

Electroporation Efficiency 10 (4) transformants / µg DNA

Per Cent Survival Not given

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Survey Number

075